

CariSurg MedTech Pathways Research Project

Preliminary Proposal

Project Title: Explainable AI Triage Assistant for Caribbean Emergency Departments

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Statement of the Problem:

Emergency departments rely on rapid triage decisions to prioritise patient care, yet heavy workloads, patient surges and human subjectivity can contribute to inconsistent triage outcomes. Recent studies have shown that artificial intelligence can support emergency department triage by improving speed and predictive performance, but many systems operate as “black boxes,” providing little explanation for how recommendations are made (Tyler et al., 2024; Da’Costa et al., 2025). This lack of explainability can make AI recommendations difficult for clinicians to understand, verify and confidently incorporate into their decision-making processes, limiting their practical use in healthcare settings (Bhuvaneswari et al., 2025).

Previous Work:

Recent research demonstrates that artificial intelligence can improve the speed, consistency and accuracy of emergency department triage. Tyler et al. (2024), Da’Costa et al. (2025) and Almulihi et al. (2024) reported that machine learning systems frequently outperform traditional triage approaches and can support patient prioritisation, admission prediction and resource planning. Bhuvaneswari et al. (2025) highlighted the importance of human-centred AI in resource-constrained settings while Kim et al. (2024) demonstrated the potential of AI to extract clinically relevant information from triage notes. Huang et al. (2022) further emphasised the need to evaluate fairness and bias in clinical machine learning systems. Several studies [Araouchi and Adda (2024), Agius et al. (2025), Tschoellitsch et al. (2023)] also reported that models were typically developed and tested within a single institution, limiting evidence of performance in other healthcare settings. Collectively, these studies suggest that limited explainability, concerns about clinician trust and uncertainty about performance across healthcare settings remain important barriers to implementation.

Proposed Solution:

Develop and evaluate an AI decision-support prototype for emergency department triage using a de-identified dataset. In addition to predicting patient triage levels, the system will provide a clear explanation of the clinical factors contributing to each recommendation. The project will evaluate whether an explainable AI model can achieve triage classification performance comparable to published studies while displaying the factors influencing each prediction. The project will use publicly available emergency department triage data and is intended as an initial step toward explainable triage tools suitable for resource-constrained Caribbean healthcare settings.

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